Introduction

With the rapid development of technology and online services food delivery applications have become an essential part of daily life Customers now expect fast easy and reliable ways to order food from different restaurants using their smartphones or computers However traditional and semi manual food delivery systems often face problems such as delayed orders communication issues and inefficient order management especially during busy hours

This project presents an Online Food Delivery System OFDS designed to improve the overall food ordering and delivery process by connecting customers restaurants delivery drivers and system administrators through one integrated platform The system allows customers to browse restaurant menus place orders make online payments track deliveries in real time and submit ratings and reviews Restaurants can efficiently manage menus and incoming orders while delivery drivers can receive delivery assignments and update delivery status instantly

The main goal of this project is to develop a secure reliable and user friendly system that enhances customer satisfaction improves communication between all users and increases operational efficiency The proposed system also supports scalability high performance and real time updates making it suitable for modern food delivery services and future business growth

Problem Description

The current food delivery system is semi-manual, which leads to delays, errors, and inefficiency, especially during peak hours. Customers experience slow service and difficulty tracking their orders, while restaurants struggle with managing large numbers of incoming requests. Delivery drivers may also face communication problems when updating delivery status. Therefore, an automated food delivery system is needed to improve efficiency, reduce errors, speed up order processing, and provide a better user experience for customers, restaurants, and delivery drivers.

Stakeholder Identification

- **Customer:** A user who orders meals through the system, tracks orders, and provides ratings or feedback.
- **Delivery Driver:** Responsible for delivering orders to customers and updating delivery status in the system.
- **Restaurant Staff:** Responsible for preparing meals, managing incoming orders, and updating order preparation status.
- **Restaurant Manager:** Oversees restaurant operations, monitors order processing, and ensures service quality..
- **System Administrator:** Manages the system, user accounts, monitors platform performance, and handles technical issues or complaints.

System Interaction

- **Customer:** Browses restaurants and menus, places orders, makes payments, tracks order status, and may cancel orders or provide ratings and reviews.
- **Restaurant:** Receives orders, manages menu items, prepares food according to requests, and updates order status.
- **Delivery Driver:** Accepts delivery tasks, delivers orders to customers, and updates delivery progress in the system.
- **System Administrator:** Monitors system performance, manages user accounts, handles complaints, and analyzes system data to improve efficiency and service quality.

Functional Requirements

1. **User Registration and Login:** The system shall allow users to register by providing personal details and securely log in using authentication credentials.

2. **Browse Restaurants and Menu:** The system shall allow customers to view a list of restaurants and browse menus. Customers shall be able to filter restaurants based on cuisine type, price, ratings, or delivery time.

3. **Order Food:** The system shall allow customers to select items, place orders, edit cart items, and view the total cost including delivery fees before confirming the order.

4. **Order Tracking:** The system shall provide real-time tracking of orders with statuses such as: order confirmed, preparing, picked up, on the way, and delivered.

5. **Payment Processing:** The system shall support multiple payment methods such as credit cards, digital wallets, and cash on delivery. The system shall confirm successful payments and generate receipts.

6. **Order Cancellation and Ratings:** The system shall allow customers to cancel orders (based on status) and provide ratings and reviews for restaurants and delivery service.

Non-Functional Requirement

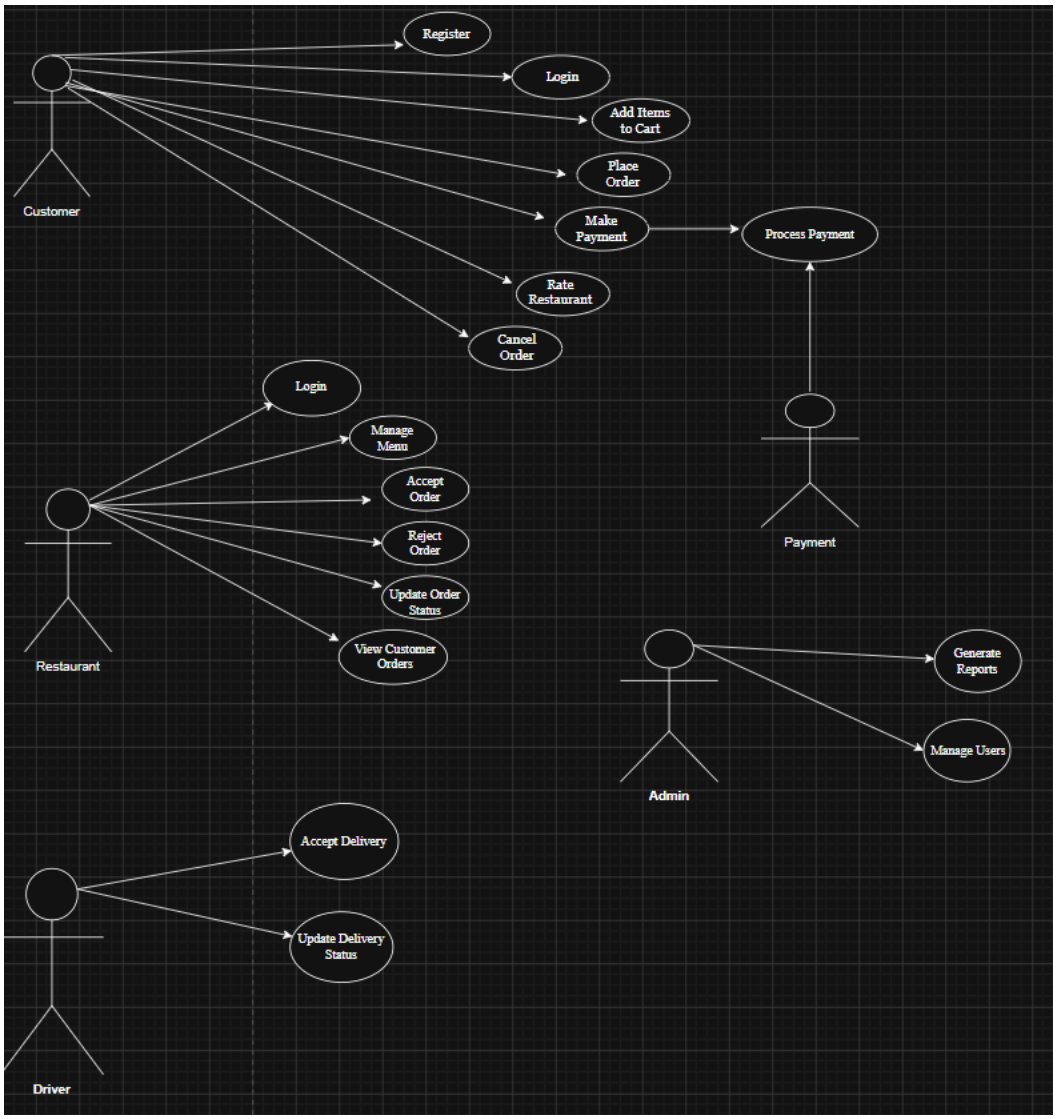
1. **Performance:** The system should respond quickly and handle a large number of orders efficiently, especially during peak hours.

2. **Security:** The system must protect user data and payment information using encryption and secure authentication methods.

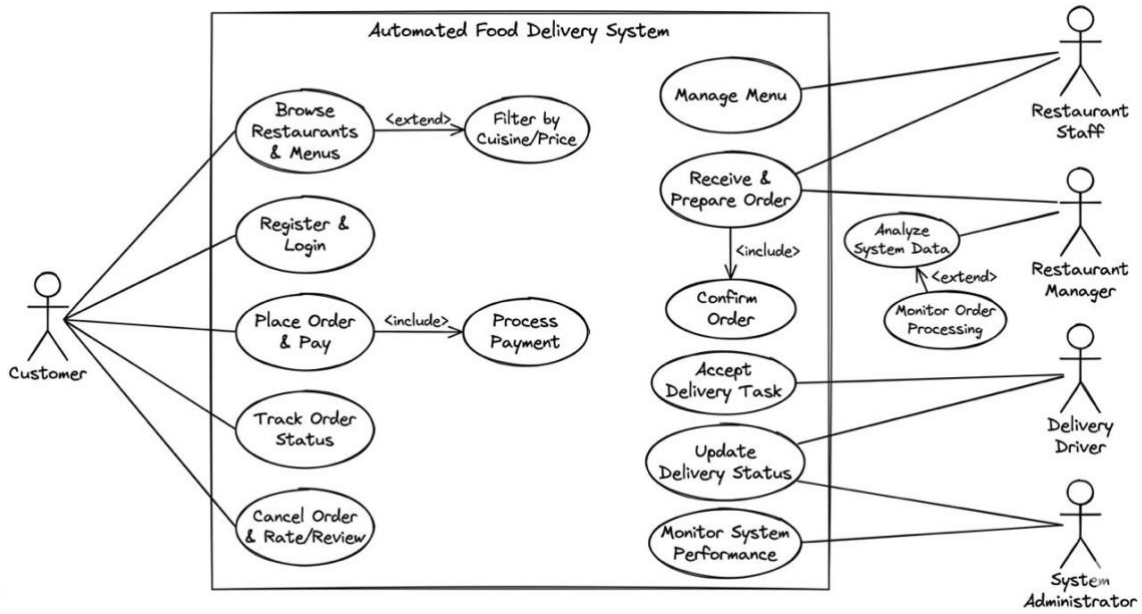
3. **Usability:** The system should have a simple, user-friendly, and intuitive interface for all users.

4. **Reliability & Availability:** The system should be highly reliable and available 24/7 with minimal downtime.

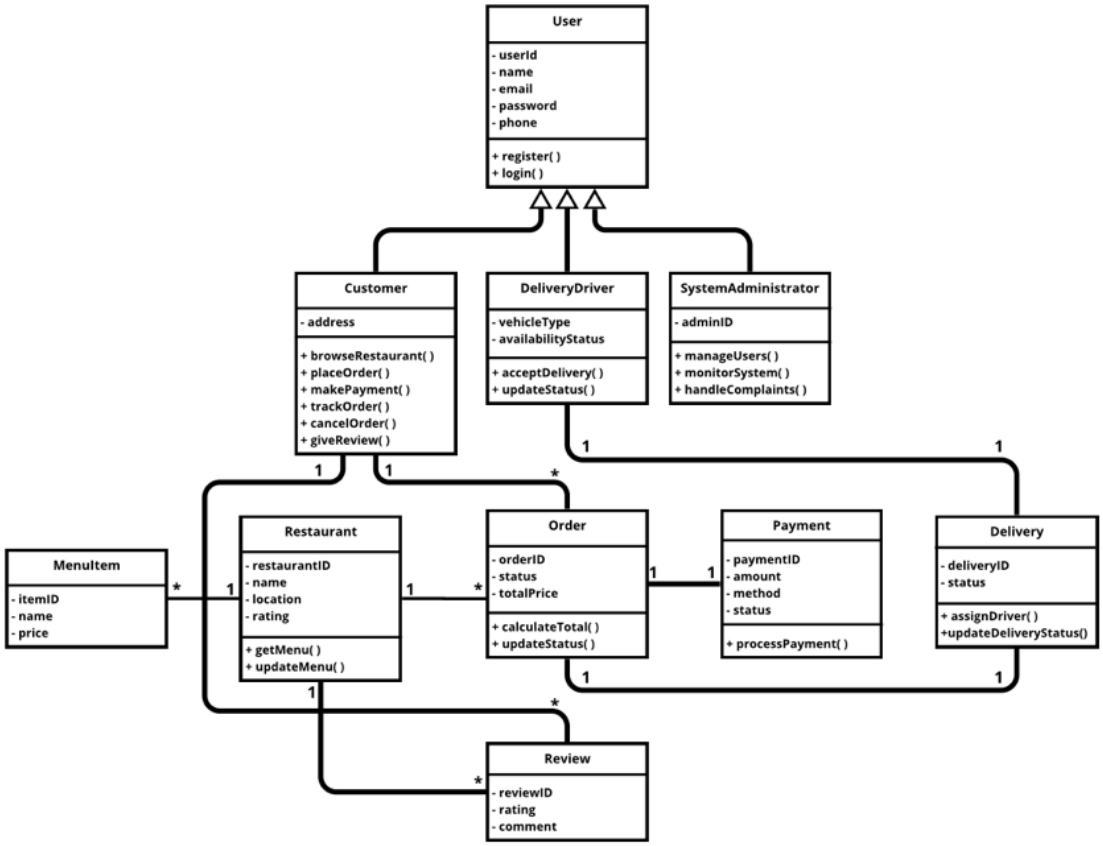
Use Case Diagram



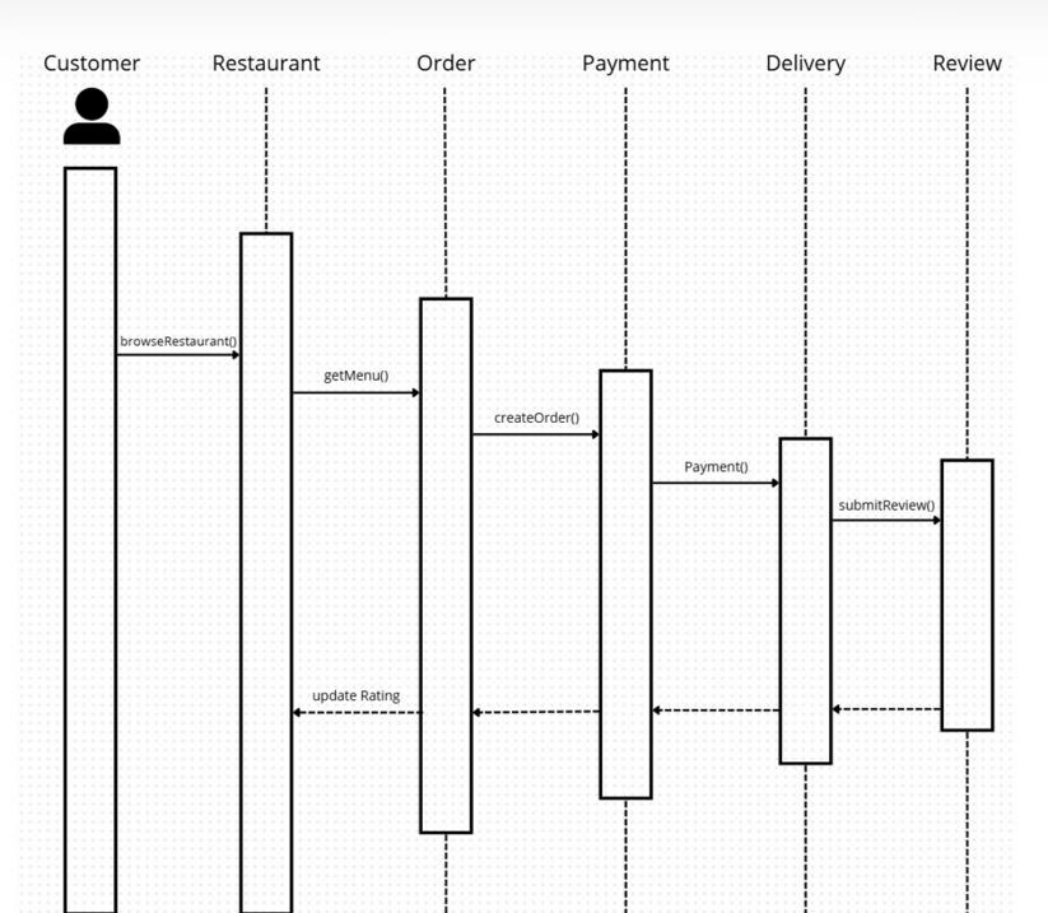
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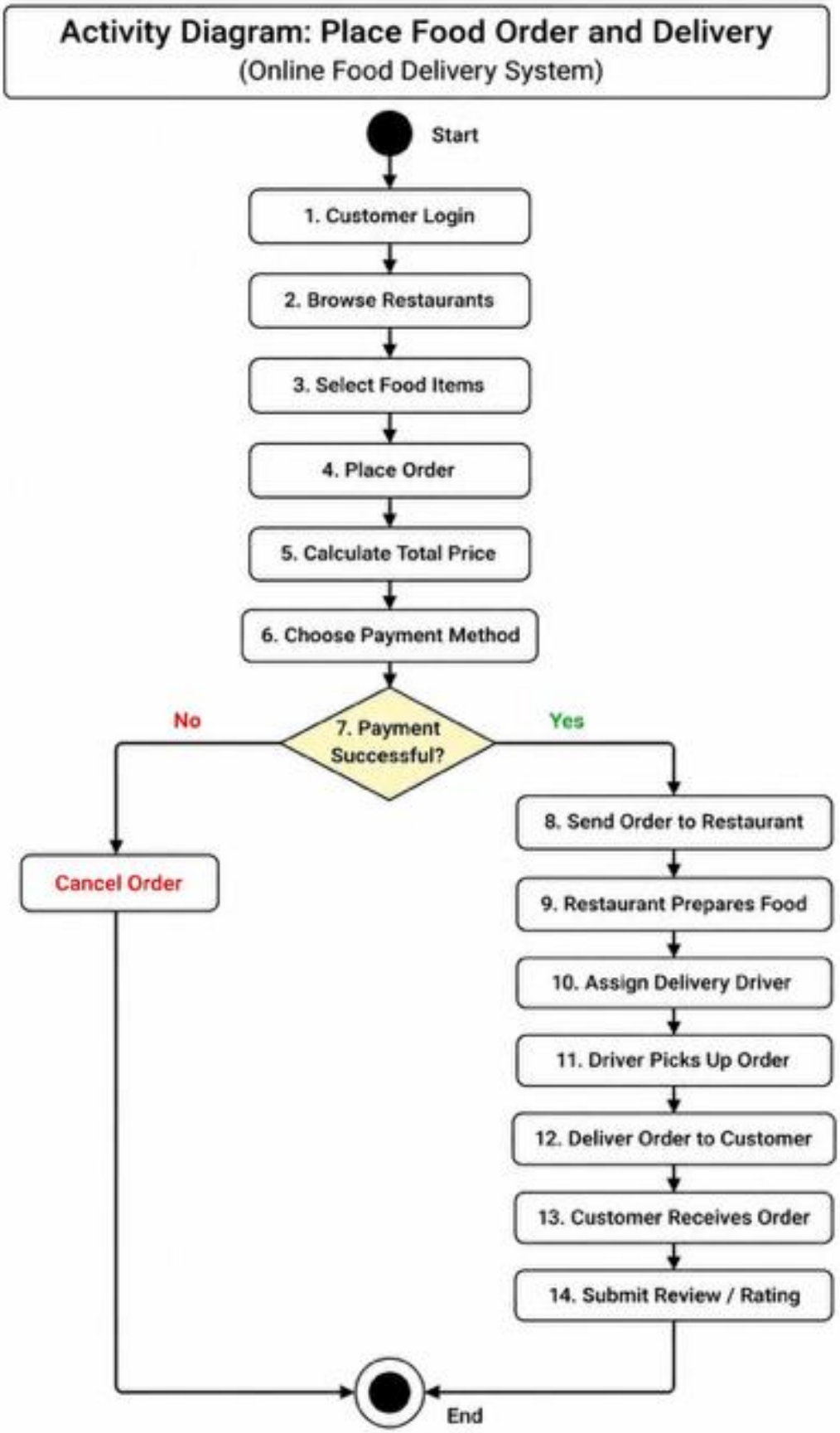
Class Diagram



Sequence Diagram



Activity Diagram



Conclusion

In conclusion the Online Food Delivery System OFDS provides an effective solution for improving the food ordering and delivery process by automating important operations and reducing delays and errors The system enhances communication between customers restaurants delivery drivers and administrators through one centralized platform

The proposed system offers many useful features including restaurant browsing online ordering secure payment real time order tracking and delivery management These features help improve customer satisfaction increase operational efficiency and provide a better overall user experience

Furthermore the system is designed to be secure reliable scalable and easy to use which makes it suitable for handling large numbers of users and orders especially during peak hours Overall this project demonstrates how software engineering concepts and system design techniques can be applied to develop a modern and efficient online food delivery solution